

Ethamukkala Karthik

Full-Stack Developer

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PROFESSIONAL SUMMARY

Aspiring Software Engineer and Full-Stack Developer specializing in Cloud & Edge Computing, with hands-on experience in building and deploying scalable web applications. Experienced in Java, Python, React, FastAPI, JavaScript, SQL, and AWS cloud services, with strong foundations in backend systems, API integration, and cloud deployment. Successfully developed production-grade projects, including a QR-based attendance system, demonstrating expertise in software development, problem-solving, and modern cloud technologies.

EDUCATION

KL University, B.Tech – Computer Science & Engineering	2023 - 2027
VINJEE Junior College, Intermediate (MPC)	2021 - 2023

TECHNICAL SKILLS

Languages: Java, Python, C, JavaScript, TypeScript

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS, shadcn/ui, Responsive Web Design

Backend: FastAPI, Node.js, Express.js, Spring Boot, RESTful APIs, Microservices Architecture, JWT

Databases: PostgreSQL, MySQL, MongoDB, Redis — Relational and Non-relational

Cloud & DevOps: AWS (EC2, S3, IAM, RDS), Microsoft Azure, Vercel, Render, Git, GitHub

Tools: VS Code, Postman, pgAdmin, Swagger UI

CS Fundamentals: OOP, Data Structures & Algorithms, MVC Architecture, SDLC, System Design basics

PROJECT EXPERIENCE

QRatten — Cloud-Based QR Attendance Platform

Next.js 15, React 19, Python, TypeScript, Tailwind CSS, FastAPI, PostgreSQL, Redis, JWT, Vercel, Render

- Eliminated proxy attendance entirely by engineering rotating QR tokens with **30-second refresh cycles** and device fingerprint validation across campus.
- Built **3 role-specific dashboards** (Student, Teacher, Admin) with real-time QR feed, absentee alerts, attendance streaks, and fraud analytics — reducing manual attendance work by **~80%**.
- Architected and deployed solo: Next.js on Vercel + FastAPI/PostgreSQL/Redis on Render, achieving **sub-200ms average API response time**.

E-Learning Quiz Application

React.js, Node.js, Express.js, MongoDB, PostgreSQL, REST APIs

- Developed a full-stack quiz platform supporting **50+ concurrent users** with real-time scoring, JWT auth, and a hybrid MongoDB + PostgreSQL data layer for flexible content storage.
- Designed RESTful APIs handling quiz creation, submission, and leaderboard endpoints with **sub-100ms query performance** on indexed PostgreSQL tables.

Real-Time Chat Application — Java Sockets

Java, Java Sockets, MongoDB, HTML, CSS

- Built a multi-user real-time chat system using Java Socket architecture, supporting **20+ simultaneous connections** with persistent message history stored in MongoDB.

CERTIFICATIONS

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| • AWS Certified Cloud Practitioner | • AWS Solutions Architect – Associate |
| • Certified Advanced Automation Professional | • HTML and CSS in Depth — Coursera |
| • Java Database Introduction — Coursera | • Operating Systems — Coursera |

ACHIEVEMENTS & ACTIVITIES

- Actively solved competitive coding problems on **LeetCode** and **CodeChef** across Data Structures, Algorithms, and Dynamic Programming.
- Competed in **4+ college-level hackathons**, consistently delivering working full-stack prototypes within 8 - 12 hours.
- Demonstrated strong ownership, teamwork, and time management skills through hackathons, project development, and independent learning initiatives.
- Developed hands-on expertise in full-stack application development, scalable backend systems, and deployment through real-world project implementation.